

Charilaos I. Kanatsoulis

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RESEARCH INTERESTS

Foundation models for structured data (relational databases, graphs, single-cell data, protein sequences) · Multi-modal language and knowledge-graph models · Structured large model pretraining · Relational AI-agents · Parameter-efficient adaptation of large pretrained models · Graph representation learning.

EDUCATION

Ph.D., Electrical & Computer Engineering – University of Minnesota, Twin Cities 2020
Thesis: Tensor Methods for Signal Reconstruction and Network Embedding. Advisor: Prof. Nikolaos Sidiropoulos.

Diploma (Dipl.-Ing.), Electrical & Computer Engineering – National Technical University of Athens 2014

EXPERIENCE

Stanford University Stanford, CA
Research Scientist, SNAP Group (PI: Jure Leskovec) 2024 – present

Leading ML research at the SNAP Group. Designing foundation models for forecasting over knowledge bases and relational databases; graph transformer architectures and positional encodings; large-scale benchmarks for relational deep learning. Developing text-to-knowledge-graph extraction methods with LLMs and transformer architectures for cell biology and protein modeling. Industry-academic collaborations with NVIDIA, SAP, Kumo.AI, and the Chan Zuckerberg Initiative. Instructor for CS224W (Machine Learning with Graphs) and CS246 (Mining Massive Datasets).

University of Pennsylvania Philadelphia, PA
Postdoctoral Researcher, Alelab (PI: Alejandro Ribeiro) 2021 – 2023

Led work on GNN design and expressivity. Developed parameter-efficient tensor adaptation methods for LLMs – reducing trainable parameters by up to two orders of magnitude vs. LoRA. Designed decentralized GNN policies for multi-agent coordination. Instructor for ESE5140 *Graph Neural Networks*.

Sentera, Inc. Minneapolis, MN
Research Scientist 2020 – 2021

Designed and benchmarked deep-learning super-resolution pipelines for agricultural imaging (UAV/satellite crop-monitoring cameras); shipped models to the production imaging stack.

SELECTED PUBLICATIONS

FULL LIST (40+) ON GOOGLE SCHOLAR

- [1] R. Ranjan, V. Hudovernik, M. Znidar, **C. I. Kanatsoulis**, et al., J. Leskovec. “Relational Transformer: Toward Zero-Shot Foundation Models for Relational Data.” **ICLR 2026**; **Oral** at EurIPS Workshop on AI for Tabular Data.
- [2] V. P. Dwivedi, S. Jaladi, Y. Shen, F. López, **C. I. Kanatsoulis**, R. Puri, M. Fey, J. Leskovec. “Relational Graph Transformer.” **ICLR 2026**; **Best Paper Award** at KDD Temporal Graph Learning Workshop.
- [3] B. Mo, K. Yu, J. Kazdan, et al., **C. I. Kanatsoulis**, S. Koyejo. “KGGGen: Extracting Knowledge Graphs from Plain Text with Language Models.” **NeurIPS 2025**.
- [4] T. Chen, **C. I. Kanatsoulis**, J. Leskovec. “RelGNN: Composite Message Passing for Relational Deep Learning.” **ICML 2025**.
- [5] **C. I. Kanatsoulis**, E. Choi, S. Jegelka, J. Leskovec*, A. Ribeiro*. “Learning Efficient Positional Encodings with Graph Neural Networks.” **ICLR 2025**.
- [6] **C. I. Kanatsoulis**, A. Ribeiro. “Counting Graph Substructures with Graph Neural Networks.” **ICLR 2024**.
- [7] I. Hounie, **C. I. Kanatsoulis**, A. Tandon, A. Ribeiro. “LoRTA: Low Rank Tensor Adaptation of Large Language Models.” *Under review*.

SELECTED PROJECTS

Relational Deep Learning & RelBench (*collaboration with NVIDIA, SAP, Kumo.AI*) – foundation-model architectures and community-standard benchmark for forecasting over relational databases; **ICLR 2025**, **ICML 2025**, **ICLR 2026**.

LoRTA (*collaboration with Microsoft*) – low-rank tensor adaptation for LLMs; up to two orders of magnitude fewer trainable parameters vs. LoRA at matched performance on DPO, instruction tuning, and protein folding.

GREmLN (*collaboration with Chan Zuckerberg Initiative*) – multimodal transformer architecture trained with single-cell transcriptomics and gene-regulatory-network structure.

KGGen (*NeurIPS 2025*) – text-to-knowledge-graph extraction with LLMs; **1.1k+** **GitHub stars**.

OPEN SOURCE

KGGen (*NeurIPS 2025*) – text-to-knowledge-graph extraction with LLMs; **1.1k+** **GitHub stars**.

RelBench v2 – community-standard benchmark and data repository for relational deep learning; adopted across SAP and Kumo.AI teams; **350+** **GitHub stars**.

Relational Transformer (*ICLR 2026* reference implementation) · **RelGNN** (*ICML 2025*) · **Pearl** (*ICLR 2025* – efficient positional encodings for GNNs) · **LoRTA** (low-rank tensor adaptation for LLMs).

SELECTED AWARDS

Best Paper Award – KDD Temporal Graph Learning Workshop (2025), *Relational Graph Transformer* · **Best Student Paper** – IEEE CAMSAP (2023) · Best Student Paper Finalist – IEEE Asilomar (2023) · ADC Graduate Fellowship in Wireless & Networking Technology, U. Minnesota (2014).

INVITED TALKS (SELECTED)

2026: UC Santa Cruz · **2025:** Google Research, UC Berkeley, UC San Diego, Stanford · **2024:** Caltech, Stanford · **2023:** UC San Diego, Stanford, Ecole Polytechnique (Paris), TU Darmstadt, University of Maryland, Rensselaer Polytechnic Institute · **2020:** Amazon Alexa, USC · **2019:** Baidu · **2018:** Arm Inc.

WORKSHOP & TUTORIAL ORGANIZATION

ICML 2026 (Graph Foundation Models) · **NeurIPS 2025** (New Perspectives in Graph ML) · LoG 2025 (Relational Deep Learning) · **KDD 2025** (Relational Deep Learning Tutorial) · Stanford Graph Learning Workshop (2024, 2025) · **ICASSP 2023** (Graph Neural Networks Tutorial).

SERVICE

Area Chair – AAAI, KDD, ICML Workshops. Reviewer – NeurIPS, ICLR, ICML, KDD, TSP.

SKILLS

Programming & tooling: Python primary (PyTorch, PyTorch Lightning, PyG, DGL, NetworkX, scikit-learn, NumPy/SciPy, Pandas); JAX/Flax familiar; MATLAB (CVX, Yalmip); distributed training (DDP, FSDP) for large-scale graph and LLM workloads.

Languages: Greek (native), English (professional).